RANDOM GRAPHS OVER BOUNDED LATTICE REGIONS

Phillip Korolev

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Abstract

We define a finite and bounded 2-dimensional lattice graph of radius r, explore general graphs constructed over the bounded set $\Lambda_r = \{x \in \mathbb{Z}^2 : \|x\|_\infty \leq r\}$, and explore randomized graph construction over this space.

1 The Bounded Lattice Graph

The 2-dimensional integer lattice is the set $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$. From this set, we can build the infinite lattice graph $\mathbb{L}^2$ with vertices $V(\mathbb{L}^2) = \mathbb{Z}^2$ and edges connecting any two vertices u, v so long as $d(u, v) = 1$ with respect to the standard metric. Now, we define the finite, bounded subgraph $\mathbb{L}^2(r) \subset \mathbb{L}^2$:

$$\mathbb{L}^2(r) = \mathbb{L}^2\left[\{v \in \mathbb{Z}^2 : \|v\|_\infty \leq r\}\right], r \in \mathbb{Z}^+ \tag{1}$$

We may choose to call this the lattice graph of radius r, and we may think of its vertex set $V(\mathbb{L}^2(r))$ as the standard Λ_r: the collection of points $(x, y) \in \mathbb{Z}^2$ for which $x, y \in [-r, r]$. The geometry of the vertex set alone is that of a discrete box of radius r bounded by the $\mathbf{L}^\infty$ norm. For the entire graph, however, we use the standard notion of distance and edge length/weight implied on $\mathbb{L}^2$.

From this initial definition, we can immediately see that $\mathbb{L}^2(r)$ is finite and bounded by $\mathbf{L}^1$ metric. As for the vertex set, it is clearly finite with $|V(\mathbb{L}^2(r))| = |\Lambda_r| = (2r+1)^2 < \infty$. We know that any node $u \in \mathbb{L}^2$ has degree 4. In $\mathbb{L}^2(r)$, a node's degree is always less than or equal to 4, depending on if it lies on the edge or center of the graph, thus the edge set $E(\mathbb{L}^2(r))$ must be finite as well:

$$|E(\mathbb{L}^2(r))| < \max_{u \in V_r} \deg(u) \cdot |V_r| \leq 4|\Lambda_r| < \infty$$

As for the boundedness, the maximum path distance between any two vertices in $\mathbb{L}^2(r)$ is achieved when they lie on opposing corners of the graph. In this case, for any $u, v \in \mathbb{L}^2(r)$,

$$d(u, v) \leq d_{\mathbf{L}^1}((-r, -r), (r, r)) = 4r < \infty$$

2 General Graphs over Λ_r

From $\mathbb{L}^2(r)$, we can use the vertex domain $\Lambda_r = V(\mathbb{L}^2(r))$ and the standard $\mathbf{L}^1$ distance to construct a general bounded graph *over* Λ_r. We can define a weighted graph over Λ_r as $G = (V, E)$, where $V \subseteq \Lambda_r$ and for all edges $e = (u, v) \in E$, $l(e) = d_{\mathbf{L}^1}(u, v)$. For shorthand notation, we may also write $G(\Lambda_r)$ to refer to a graph generated over Λ_r.

As the space Λ_r is bounded, it follows that any graph over, or embedded in, Λ_r must also be bounded. Similarly, $|\Lambda_r|$ is finite, so $V \subseteq \Lambda_r$ is too.

2.1 Induced Metric & Inner Regions

Under the $\mathbf{L}^1$ metric, the triangle inequality still holds. That is, for any vertices $u, v, w \in V(G(\Lambda_r))$, the path distance $d(u, w) \leq d(u, v) + d(v, w)$. If vertices u, w are not parallel, and v lies inside or on the rectangular box with u, w as opposing corners, then $d(u, w) = d(u, v) + d(v, w)$.

We can develop a standard notion for this box inside Λ_r, with vertices u, v as opposing corners. We will call it the rectangular region encapsulated by opposing corners u and v, denoted $R(u, v) \subseteq \Lambda_r$. For any $u, v \in V(G(\Lambda_r))$:

$$R(u, v) = \{x \in \Lambda_r : d(u, v) = d(u, x) + d(x, v)\},$$

with respect to the standard $\mathbf{L}^1$ metric. It not need be that x is a vertex in $G(\Lambda_r)$, this is simply a definition of a region in Λ_r.

For any integer grid Λ_r, we can denote the (global) corners as quadrants of the xy axis. Beginning at the top-right corner and moving counter-clockwise, we have corners C_1, C_2, C_3, $C_4 \in \Lambda_r$. If we build the rectangular region encapsulted by two opposing global corners, then this region makes up all of Λ_r. That is,

$$R(C_a, C_{(a+2) \bmod 4}) = \Lambda_r$$

2.2 Isomorphisms & Embeddings

Let H be any abstract graph that is not necessarily embedded in Λ_r. There are particular conditions that must be satisfied for H to be represented as a isomorphic graph embedded in some Λ_r. Assume $H = (V, E)$ is any abstract *unweighted* graph. In this case, edge length is negligible, so as long as we have a set of vertices $V' \subseteq \Lambda_r$ with $|V'| \geq |V|$, then we can build a graph $G(\Lambda_r)$, that is isomorphic to H. As long as $|V(G(\Lambda_r))| \geq |V(H)|$, there must exist an injective mapping $\phi : V(H) \to V(G(\Lambda_r))$. As for the edges, if $(u, v) \in E(H)$, we simply reconnect the images of these vertices in $G(\Lambda_r)$ to achieve the isomorphic graph:

$$(u, v) \in E(H) \implies (\phi(u), \phi(v)) \in E(G(\Lambda_r)) \tag{2}$$

We know that $|\Lambda_r| = (2r + 1)^2$. Thus, for any unweighted graph H with n vertices, we require $r \geq (\lceil(\sqrt{n} - 1)/2\rceil)$ to embed it into some finite, bounded lattice space Λ_r.

Theorem. For all unweighted $H = (V, E)$ with $|V| = n$, there exists a graph $G(\Lambda_r)$ with $r \geq \lceil(\sqrt{n} - 1)/2\rceil$ such that $H \cong G(\Lambda_r)$.

This process only fails if we wish to embed an infinite graph K with $|V(K)| = \infty$. In this case, we could only embed onto $\Lambda_\infty = \mathbb{Z}^2$.

Now, what if we want to embed a *weighted* graph $H = (V, E, w)$, with weight function $w : E \to \mathbb{Z}^+$, into some Λ_r? From the above, we know how minimal r can be, but if we only considered the size of the vertex sets, then the case where $\max_{e \in E(H)} w(e) > 4r$ would make an embedding into what we thought we required for r impossible. Thus, there are additional constraints we need to cover when dealing with weighted graphs in order to guarantee the existence of an isomorphic counterpart $G(\Lambda_r)$.

Consider a weighted graph $H = (V, E, w)$. For reference, a graph embedded in Λ_r has weight function $w(e) = \|e\|_{\mathbf{L}^1}$. For starters, let us assume that H has only 2 nodes. Then, we require $r \geq \lceil(\sqrt{2} - 1)/2\rceil = 1$ to ensure we can at least map all of the nodes from H. With $r = 1$, we know the diameter of Λ_r, and equivalently, the maximum length of an edge from $G(\Lambda_r)$, is $4r = 4$. However, if the edge in H connecting its only two nodes has weight $w(e) > 4$, we cannot embed H in Λ_1. So, we not only need $|\Lambda_r| \geq |V(H)|$, but we also require:

$$\max_{e \in E(H)} w_H(e) \leq 4r \implies r \geq \frac{1}{4} \max_{e \in E(H)} w_H(e)$$

Thus, when finding the condition on r that would ensure the existence of a graph $G(\Lambda_r)$ giving the isomorphism $H \cong G(\Lambda_r)$ for a weighted graph H, we should first choose r as described above, then increase it as needed to ensure we can sufficiently map all vertices $V(H)$.

Consider some graph $G(\Lambda_R)$, what type of conditions should hold in order to guarantee the existence of an isomorphic $G(\Lambda_r)$ with $r < R$? Without taking into account edge weights, we can guarantee the existence of an isomorphic underlying graph $G(\Lambda_r)$ if and only if

$$\text{(i) } R > \lceil (\sqrt{V(G(\Lambda_R))} - 1)/2 \rceil$$

With this condition satisfied, we can at least guarantee an injective vertex mapping $\phi : V(G(\Lambda_R)) \to \Lambda_r$. Now, if we want an strict graph isomorphism $G(\Lambda_R) \cong G(\Lambda_r)$, then we must have (i), but also

$$\text{(ii) } r \geq \frac{1}{4} \max_{e \in E(G(\Lambda_R))} w(e)$$

3 Random Graphs Over Λ_r

We begin the section by building a notion of randomized graph embedded in Λ_r. We must choose a random process that constructs both vertex and edge sets. Let us define the random graph over Λ_r as the extended notation $G(\Lambda_r; p_v, p_e) = (V, E)$ with conditions:

- $u \in \Lambda_r$ is apart of V with probability p_v.
- an edge exists between any two vertices $u, v \in V$ with probability p_e.

3.1 Basic Properties

From the random construction described above, the expected number of vertices n is:

$$\mathbb{E}[n] = p_v |\Lambda_r|$$

As for the edges, let us first look at the expected degree of a vertex $u \in V(G(\Lambda_r; p_v, p_e))$. For any such vertex, there are $p_v|\Lambda_r| - 1$ choices for connection by the above. As an edge exists between any two vertices with probability p_e, we have:

$$\mathbb{E}[\deg(u)] = p_e(p_v|\Lambda_r| - 1)$$

Similar to the way we derive the expected number of vertices, we can consider the set of all possible edges, and multiply it by the probability p_e to achieve the expected number of edges in the graph $G(\Lambda_r; p_v, p_e)$. For any graph with n vertices, the number of possible edges without duplication or self loops is $n(n-1)/2$. Let n be our expected number of vertices as previously described, then the expected number of edges must be:

$$\begin{aligned}
\mathbb{E}\left[|E|\right] &= p_e \left(\frac{n(n-1)}{2} \right) \\
&= p_e \left(\frac{(p_v|\Lambda_r|)(p_v|\Lambda_r| - 1)}{2} \right) \\
&= p_e \left(\frac{(p_v|\Lambda_r|)^2 - p_v|\Lambda_r|}{2} \right)
\end{aligned}$$

We may also simply write it in terms of expected values:

$$\mathbb{E}[|E|] = p_e \left(\frac{\mathbb{E}[|V|]^2 - \mathbb{E}[|V|]}{2} \right)$$

Some immediate algebraic results to note are:

1. As $p_v \to 1$, the size of the vertex set $|V|$ tends to $|\Lambda_r|$.
2. Fix $p_v > 0$. As $p_e \to 1$ The expected number of edges tends to $\frac{1}{2}[(p_v|\Lambda_r|)^2 - p_v|\Lambda_r|]$.
3. Fix $p_e > 0$. As $p_v \to 1$, the expected degree of a node tends to $p_e(|\Lambda_r| - 1)$.

3.2 Additional Properties

We have derived the expected number of nodes, edges, and node degree of a random graph $G(\Lambda_r; p_v, p_e)$ generated over Λ_r. We will now develop other expectations, such as the expected average, maximum, and minimum edge weight, which will help us understand embeddings of such random graphs over smaller spaces Λ_k with $k < r$.

Already, we can be assured that $1 \leq \mathbb{E}[w(e)] \leq 4r$ for any $G(\Lambda_r; p_v, p_e) = (V, E)$ as we do not allow self-loops, and $d(\Lambda_r) = \max_{x,y \in \Lambda_r} d(x, y)$ is bounded above by $4r$ with respect to the standard $\mathbf{L}^1$ metric.

3.3 Random Graphs Embedded in Λ_r

Previously, we handled a deterministic case for graph embeddings in Λ_r; for any existing abstract graph H, we concluded specific conditions on r required to embed H into Λ_r. Equivalent to embedding H into Λ_r, we can say: building a graph over Λ_r, $G(\Lambda_r)$, such that $H \cong G(\Lambda_r)$. We explore the case where we wish to embed random graphs.

Let us consider a random graph $G(n, p)$ generated by the Gilbert model. The number of vertices n is fixed, and p represents the probability that an edge exists between any pair of vertices. If $G(n, p)$ is unweighted, then recall that we can always guarantee an isomorphism $G(\Lambda_r) \cong G(n, p)$ with $r \geq \lceil(\sqrt{n} - 1)/2\rceil$. Thus, $G(n, p)$ can always be embedded into some Λ_r. Following similar logic, we deduce that any graph $G(n, M)$ generated by the Erdos-Renyi model is also embeddable in Λ_r.

Theorem. Any random graph $G(n, p)$ or $G(n, M)$ generated by the Gilbert and Erdos-Renyi models respectively, can be minimally embedded in $\Lambda_{\lceil(\sqrt{n}-1)/2\rceil}$. Equivalently, for H generated by any of the two random models described, there exists an isomorphism:

$$G(\Lambda_{\lceil(\sqrt{n}-1)/2\rceil}) \cong H$$

Now, we wish to explore more complex cases. Consider the graph $G(\Lambda_R; p_v, p_e)$, does there exist an isomorphic $G(\Lambda_r; p_v, p_e)$ with $r < R$?

4 Embedding Random Walks in Λ_r